

CARPORT A — 24' × 25' TWO-CAR (CLEAR SPAN)

Qty	Member	Section	Length
4	Girders (24' clear span @ 8'-4" o.c.)	3 × 12 × .188 6005A-T61 / 6061-T6	24'
3	Side beams (two 25' runs, splices on posts)	2 × 6 × 1/8 6061-T6	24'
4	Posts (8 pcs @ ~10', two per tube)	4 × 4 × 1/4 6061-T6	24'
9	Roof deck	1-1/2" Type B, 18 GA galvanized	27'

CARPORT B — 16' × 20' TRACTOR

Qty	Member	Section	Length
6	Rafters @ 4'-0" o.c.	2 × 8 × 1/8 6061-T6	18' (cut from 20')
2	Side beams (20' runs)	3 × 6 × 1/8 6061-T6	20'
3	Posts (6 pcs @ ~9', two per tube)	3 × 3 × 1/4 6061-T6	20'
6	Roof deck	1" Type B, 22 GA galvanized	22'

CONNECTIONS & HARDWARE (BOTH CARPORTS)

Qty	Item	Use
2 bars	Aluminum angle 3 × 3 × 1/4 6061-T6	Post saddle tabs + rafter clips (shop-cut)
1 bar	Aluminum flat bar 6" × 1/4 6061-T6	Post cap plates (saddles)
28	Bolts 1/2"-13 × 4" SS316 + nylock nuts + washers	Girder/beam to post saddles — 2 per post
1 box (250)	#14 × 1" SS self-drilling HWH	Rafter clips (4 per clip)
~750	#12-14 × 1-1/4" SS self-drilling + bonded EPDM washer	Deck to framing — every flute at perimeter/edges, per schedule in field
1 box	#10 × 3/4" SS stitch screws	Deck side laps @ 24" o.c. max
1 roll	Butyl tape 2"-3" (or bituminous paint)	Isolation: galvanized deck on aluminum flanges
56	Anchors Hilti Kwik HUS-EZ SS316 1/2" × 6"	Base plates — 4 per plate (screw-in, no expansion)

CONNECTION SYSTEM (AS DETAILED IN DRAWINGS)

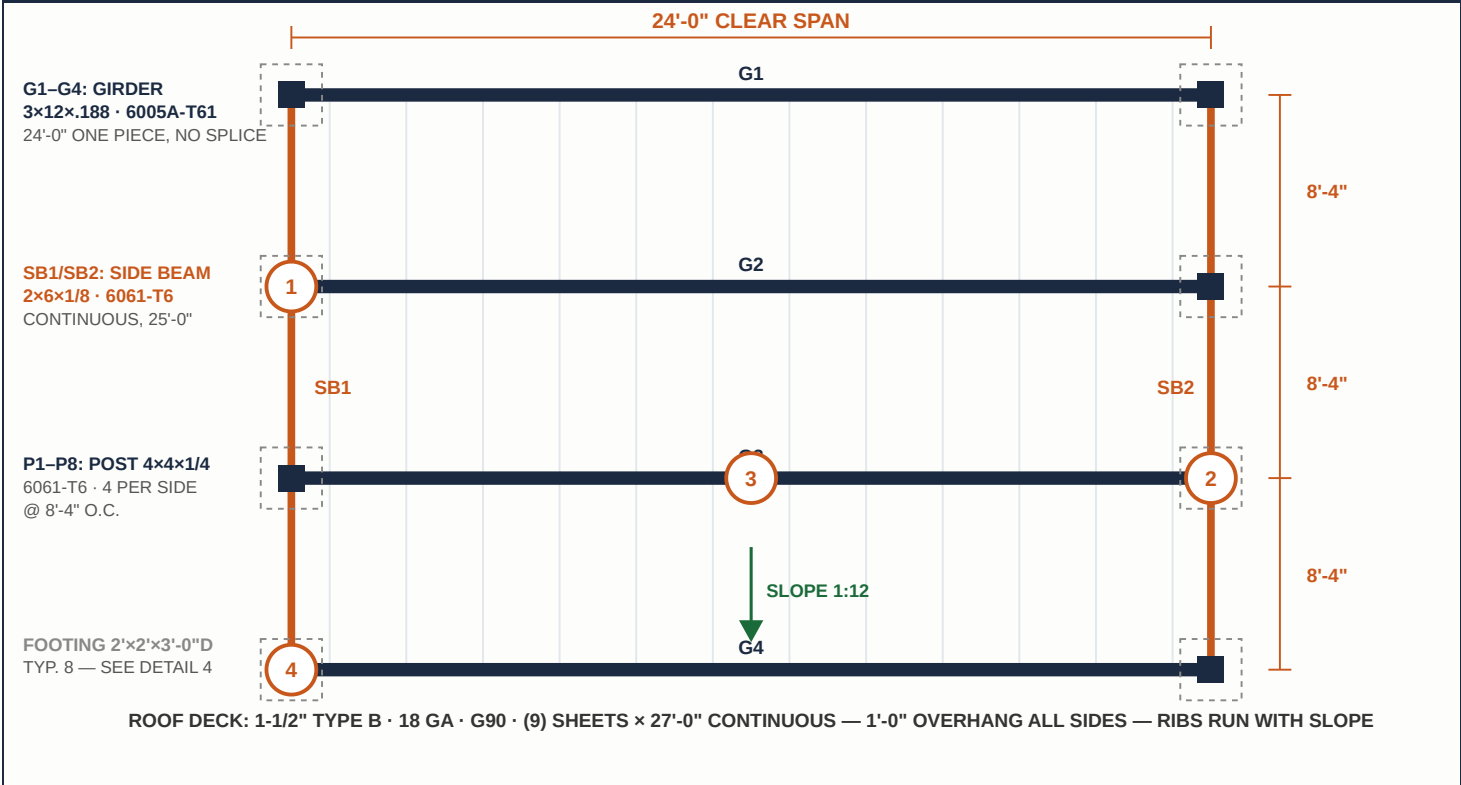
Joint	Detail
Post → girder/beam	Shop-welded saddle (1/4" cap plate + two 1/4" × 4" tabs) — beam bears on post; (2) 1/2" SS316 through-bolts take wind uplift. Field work = set & bolt only.
Rafter → side beam	Aluminum angle clip 3×3×1/4 × 3"-4", (4) #14 SS screws per clip (or one 3/8" SS bolt per leg), uphill face.
Deck → framing	SS self-drillers with EPDM washers: every flute at perimeter, ends & corners ; field per schedule. Stitch screws at side laps. Butyl isolation tape on flanges.

Post → footing	Welded aluminum base plate + gussets, (4) Hilti KWIK HUS-EZ SS316 1/2×6 into 2'×2'×3' footing, top flush, smooth & level.
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*All shop welding in 6061/6005 aluminum; field connections mechanical (stainless) only. Sizes wind-checked for 170 mph / Exposure C open structure (ASD).
Final sealed engineering by Florida P.E. governs; this sheet and the attached drawings are the fabrication basis.*

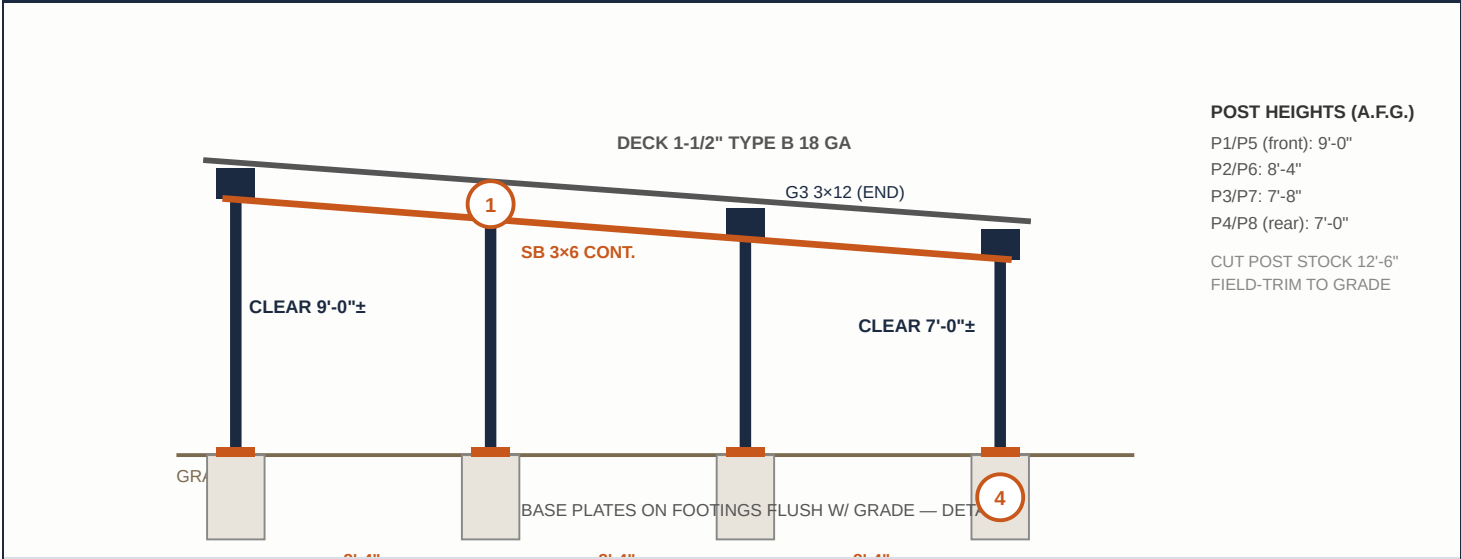
PROJECT Two-Vehicle Open Carport — 24'-0" × 25'-0"	LOCATION 18398 131st Trail N, Jupiter Farms, FL	CLIENT Raul	SCALE N.T.S.
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A-1 · FOUNDATION & ROOF FRAMING PLAN



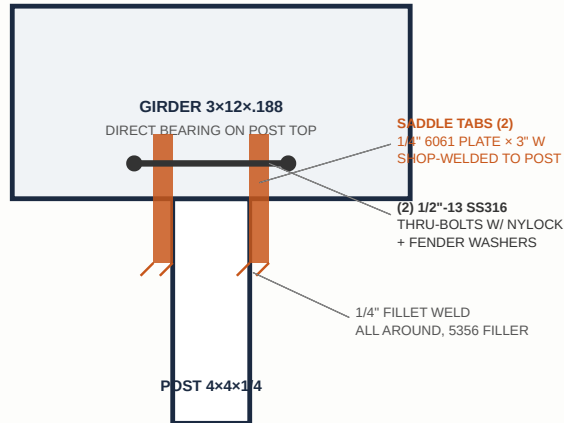
All columns on the two 25' sides only — both 12' parking bays fully unobstructed. Circled numbers reference connection details below.

A-2 · SIDE ELEVATION



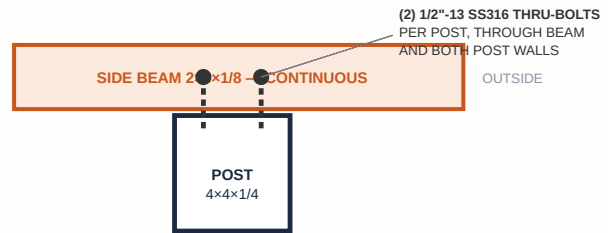
Single slope 1:12 front to rear. Girders level side-to-side; deck ribs drain front to rear.

DETAIL 1 · GIRDER-TO-POST CONNECTION



Girder bears on post top (gravity); saddle tabs + thru-bolts resist wind uplift. Typ. 8 locations.

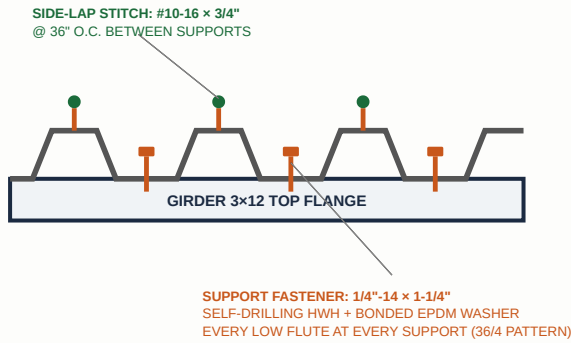
DETAIL 2 · SIDE BEAM AT POST (PLAN SECTION)



SIDE BEAM RUNS ON OUTSIDE FACE OF ALL 4 POSTS, TOP FLUSH WITH GIRDER BEARING — TIES COLUMN LINE, 25'-0" ONE PIECE (SPlice OVER POST PERMITTED, INTERNAL SLEEVE 12" + (4) 3/8" SS BOLTS)

Typ. 8 locations (4 per side). All bolt holes drilled 1/16" oversize, deburred.

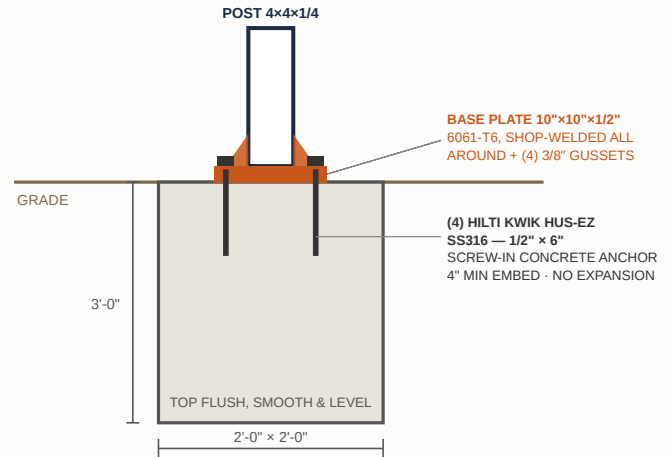
DETAIL 3 · DECK FASTENING (SECTION AT GIRDER)



DECK: 1-1/2" TYPE B · 18 GA · G90 GALVANIZED · SHEETS CONTINUOUS OVER 3 SPANS

Perimeter zone: fastener in EVERY flute. Field: every flute at supports per HVHZ practice. Butyl tape or bituminous paint between galvanized deck and aluminum flange (isolation).

DETAIL 4 · FOOTING & BASE PLATE ANCHORAGE



3,000 psi footing poured flush with grade; anchors installed after cure (min 7 days). Typ. 8.

BILL OF MATERIALS — CARPORT A

Mark	Member	Section	Alloy	Length	Qty
P1–P8	Column	4" x 4" x 1/4" sq. tube	6061-T6	12'-6" (field-trim)	8
G1–G4	Roof girder (clear span)	3" x 12" x 0.188" rect. tube	6005A-T61	24'-0"	4
SB1–SB2	Side beam (continuous)	2" x 6" x 1/8" rect. tube	6061-T6	25'-0"	2
—	Saddle tabs (Detail 1)	1/4" plate, 3" x 6"	6061-T6	—	16
—	Roof deck	1-1/2" Type B, 18 ga, G90 galv.	Steel	27'-0" (3'-0" cov.)	9
—	Footing concrete, 3,000 psi	2'-0" x 2'-0" x 3'-0"	—	0.45 CY each	8

FASTENER SCHEDULE — CARPORT A

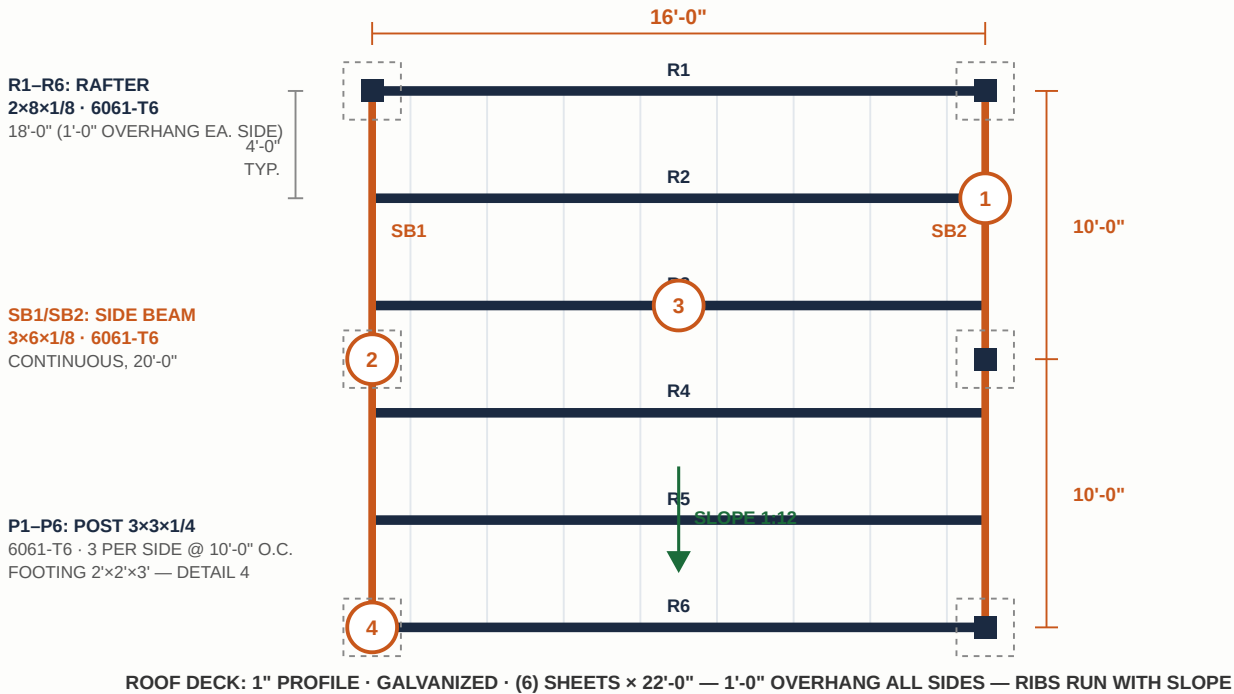
Location	Fastener	Qty / Spacing
Girder to post (Detail 1)	1/2"-13 × 5-1/2" SS316 hex bolt, nylock nut, fender washers	2 per post · 16 total
Side beam to post (Detail 2)	1/2"-13 SS316 thru-bolt, nylock nut, fender washers	2 per post · 16 total
Deck to girders (Detail 3)	1/4"-14 × 1-1/4" self-drilling HWH, bonded EPDM washer	Every low flute at every support
Deck side laps	#10-16 × 3/4" stitch screw	36" o.c. between supports
Base plate anchors (Detail 4)	Hilti KWIK HUS-EZ SS316, 1/2" × 6" screw-in concrete anchor (no expansion), 4" min embed	4 per plate · 32 total

GENERAL NOTES

- All aluminum welding with 5356 filler by certified aluminum welder; 1/4" fillet welds unless noted.
- All bolts, nuts, washers and anchor rods stainless steel Type 316. Bolt holes 1/16" oversize, deburred.
- Galvanized deck bearing directly on aluminum is a compatible (zinc) contact; all deck cut edges touched up with cold-galvanizing compound.
- Footing tops poured flush with grade, smooth and level. Anchors installed into cured concrete (3,000 psi, min 7 days) with Hilti-specified bit and torque — screw-in anchors only, no expansion anchors. Isolation barrier (bituminous paint or membrane pad) between aluminum plate and concrete.
- Layout, post locations and final grades field-verified with owner before excavation. Roof deck overhangs the frame 1'-0" on all four sides. Deck sheet lengths confirmed against as-built frame before ordering.
- Structure designed as open (no walls). Do not enclose sides without review — wind loads change.

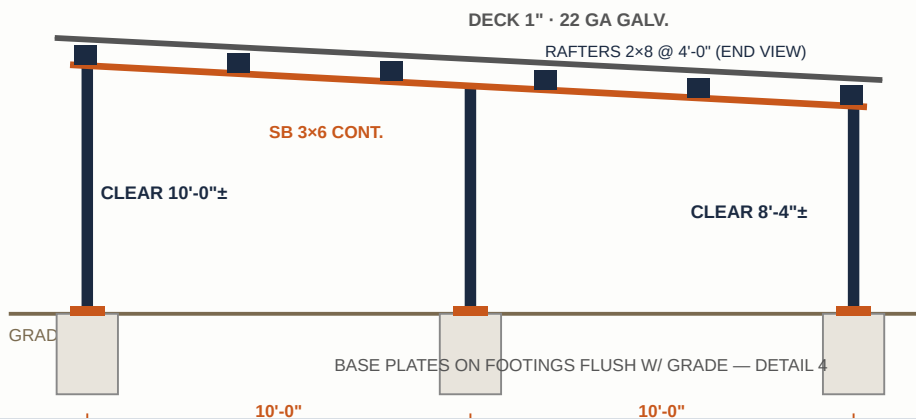
PROJECT Tractor Shelter — 16'-0" × 20'-0"	LOCATION 18398 131st Trail N, Jupiter Farms, FL	CLIENT Raul	SCALE N.T.S.
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B-1 · FOUNDATION & ROOF FRAMING PLAN



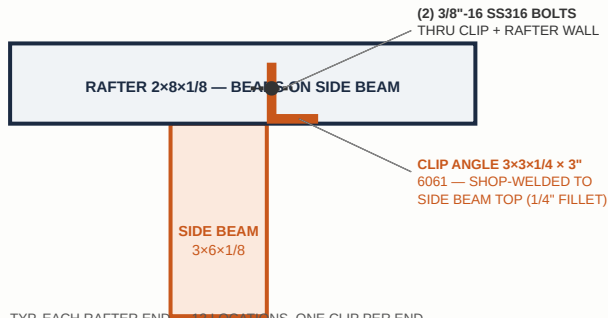
Entry at front (high) end — 10'-0" clear for tractor ROPS. Circled numbers reference details below.

B-2 · SIDE ELEVATION



Single slope 1:12 front to rear. Post heights above grade: front 10'-0", middle 9'-2", rear 8'-4" (cut from 13'-6" stock, field-trim).

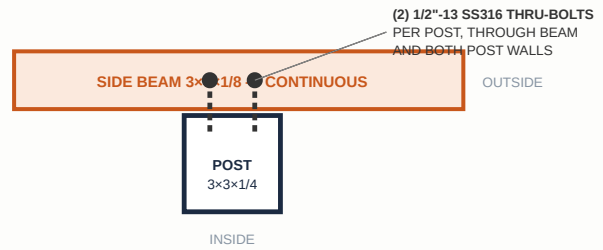
DETAIL 1 · RAFTER-TO-SIDE BEAM CONNECTION



TYP. EACH RAFTER END — 12 LOCATIONS. ONE CLIP PER END, INSTALLED ON UPHILL FACE. RAFTER OVERHANG 1'-0" BEYOND BEAM.

Gravity by direct bearing; clip + bolts resist wind uplift and alignment.

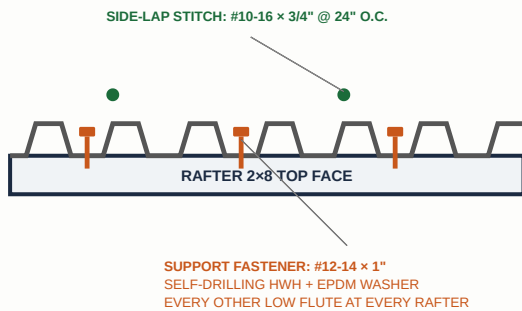
DETAIL 2 · SIDE BEAM AT POST (PLAN SECTION)



SIDE BEAM ON OUTSIDE FACE OF POSTS, TOP OF BEAM = TOP OF POST. 20'-0" ONE PIECE PER SIDE — NO SPLICES.

Typ. 6 locations. Bolt holes 1/16" oversize, deburred.

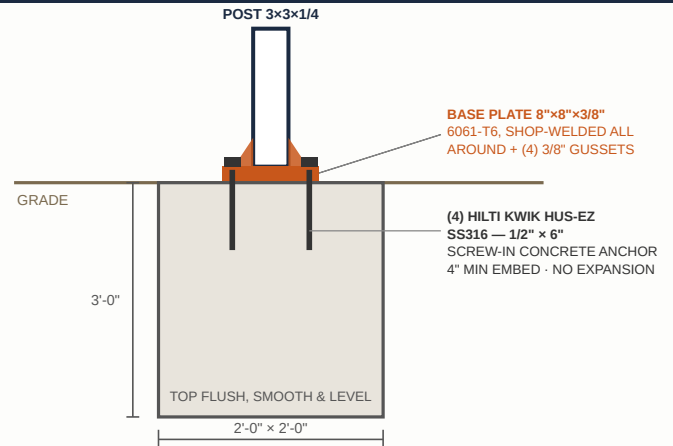
DETAIL 3 · DECK FASTENING (SECTION AT RAFTER)



DECK: 1" PROFILE · GALVANIZED · 22'-0" FULL-LENGTH SHEETS (5 SPANS)

Perimeter/edge zone: every low flute. Cut edges cold-galv touch-up.

DETAIL 4 · FOOTING & BASE PLATE ANCHORAGE



3,000 psi footing poured flush with grade; anchors installed after cure (min 7 days). Typ. 6.

BILL OF MATERIALS — CARPORT B

Mark	Member	Section	Alloy	Length	Qty
P1–P6	Column	3" × 3" × 1/4" sq. tube	6061-T6	13'-6" (field-trim)	6
R1–R6	Rafter	2" × 8" × 1/8" rect. tube	6061-T6	18'-0"	6
SB1–SB2	Side beam (continuous)	3" × 6" × 1/8" rect. tube	6061-T6	20'-0"	2
—	Clip angles (Detail 1)	3" × 3" × 1/4" angle × 3" long	6061-T6	—	12
—	Roof deck	1" profile, 20/22 ga, galvanized	Steel	22'-0" (3'-0" cov.)	6
—	Footing concrete, 3,000 psi	2'-0" × 2'-0" × 3'-0"	—	0.45 CY each	6

FASTENER SCHEDULE — CARPORT B

Location	Fastener	Qty / Spacing
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Rafter to side beam (Detail 1)	3/8"-16 SS316 bolt, nylock nut, washers, thru clip angle	2 per rafter end · 24 total
Side beam to post (Detail 2)	1/2"-13 SS316 thru-bolt, nylock nut, fender washers	2 per post · 12 total
Deck to rafters (Detail 3)	#12-14 × 1" self-drilling HWH, bonded EPDM washer	Every other low flute at every rafter; every flute at edges · butyl tape between deck and aluminum flange
Deck side laps	#10-16 × 3/4" stitch screw	24" o.c. between rafters
Base plate anchors (Detail 4)	Hilti KWIK HUS-EZ SS316, 1/2" × 6" screw-in concrete anchor (no expansion), 4" min embed	4 per plate · 24 total

GENERAL NOTES

- All aluminum welding with 5356 filler by certified aluminum welder; 1/4" fillet welds unless noted.
- All bolts, nuts, washers and anchor rods stainless steel Type 316. Bolt holes 1/16" oversize, deburred.
- Galvanized deck on aluminum is a compatible (zinc) contact; deck cut edges touched up with cold-galvanizing compound.
- Footing tops poured flush with grade, smooth and level. Anchors installed into cured concrete (3,000 psi, min 7 days) with Hilti-specified bit and torque — screw-in anchors only, no expansion anchors. Isolation barrier (bituminous paint or membrane pad) between aluminum plate and concrete.
- 10'-0" clear entry height verified against owner's tractor (ROPS raised) before fabrication.
- Layout and final grades field-verified with owner before excavation. Structure designed as open (no walls) — do not enclose without review.